

Roll No.

TME-805

**B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
ENTREPRENEURSHIP ESSENTIALS**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Define Entrepreneurship and Enlist the characteristics of Entrepreneur.

(CO1, CO2)

OR

(b) Explain any *two* ideas you wish to work on as an Entrepreneur.

(CO1, CO2)

2. (a) What is Social Entrepreneurship ? Explain.

(CO1, CO3)

OR

(b) What is the role of creativity in Entrepreneurship ? Explain.

(CO1, CO3)

3. (a) Compare Entrepreneur and an Intrapreneur.

(CO1, CO2)

OR

(b) Explain Importance of Entrepreneurship.

(CO1, CO2)

P. T. O.

(2)

4. (a) What are the challenges for an Entrepreneur in present scenario ?

(CO2, CO3)

OR

- (b) How Important is risk taking in Entrepreneurship ? Explain with an.

(CO2, CO3)

5. (a) Entrepreneurs are born on made. Comment with an example.

(CO4, CO2)

OR

- (b) Opportunities start with thousands of ideas. Explain.

(CO4, CO2)

Roll No.

UCE-801

**B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

DISASTER MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub-question among each main question.

(iii) Each question carries 10 marks.

1. (a) What is the fundamental difference between a natural disaster and a man-made disaster ? (CO1)

OR

- (b) Explain the concept of disaster risk and how it relates to vulnerability and exposure. (CO1)

2. (a) Discuss the various causative factors that can lead to the Occurrence of natural disasters. (CO2)

OR

- (b) How do human activities contribute to the occurrence and severity of certain types of disasters ? (CO2)

P. T. O.

(2)

3. (a) Explain how socioeconomic factors can influence a community's vulnerability to disasters. (CO1 & CO2)

OR

- (b) Explain emergency response plans for mitigating the impact of nuclear incidents. (CO1 & CO2)

4. (a) How can transportation safety measures be improved to reduce accident rates? (CO1 & CO2)

OR

- (b) Discuss international cooperation efforts in response to global disasters. (CO1 & CO2)

5. (a) How can preparedness strategies adapt to address new and evolving risks? (CO3)

OR

- (b) Discuss sustainable urban development strategies to address climate-related risks. (CO3)

Roll No.

TAS-801

B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

AIRPORT AND AIRLINE MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 0 marks.

1. (a) Who invented the earliest manned hot air balloon ? (CO2)

OR

(b) Who is the father of aviation. Briefly describe the contribution. (CO2)

2. (a) What are the three stages of airport planning ? (CO2)

OR

(b) Why is the aviation industry important ? (CO2)

3. (a) What is the ICAO and what are its functions ? (CO1)

OR

(b) How ICAO define 'Dangerous Goods' ? Explain. (CO1)

P. T. O.

(2)

4. (a) Show the organizational structure of an airline ? (CO1)

OR

- (b) Write main functions and responsibilities of Airports Authority of India. (CO1)

5. (a) What is the primary mission of general aviation ? (CO2)

OR

- (b) What is the difference between a Taxiway and Runway ? (CO1)

Roll No.

DEEM-06

B. TECH. (ME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

TOTAL QUALITY MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the various types of costs contributing to the cost of quality.
Give examples of each. (CO1)

OR

- (b) State and explain the dimensions of quality. Also write the principles of
Total Quality Management. (CO2)

2. (a) Explain the Taguchi's quadratic quality loss function. How does it differ
from the traditional approach of quality loss cost ? (CO2)

OR

- (b) Designer customer satisfaction questionnaire to evaluate the level of the
customer satisfaction in sport shoe manufacturer industry. (CO2)

P. T. O.

(2)

3. (a) Write short notes on the following : (CO2)

- (i) Flowcharts
- (ii) Check sheets
- (iii) Histogram

OR

(b) A welding workshops receives a lot of complaints about poor quality solder joints. Analyze this problem by constructing a cause and effect diagram. (CO2, CO3)

4. (a) Write the step by step procedure of implementing a bench marking process in an organization. (CO1, CO2)

OR

(b) Write short notes on the following : (CO2)

- (i) PDCA
- (ii) 5S
- (iii) Kaizen
- (iv) PDSA

5. (a) What is the overall effect of motivation, empowerment and performance on the total quality management ? (CO3)

OR

(b) What are the elements of ISO 9000 quality system ? Explain how quality systems are implemented, documented and audited ? (CO3)

Roll No.

DEMM-08

B. TECH. (ME) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

ADVANCED WELDING TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define welding and its classification. What are its advantages and limitations ? (CO1)

OR

- (b) Briefly classify the process selection criteria of welding process. (CO1)

2. (a) Explain oxy-acetylene gas welding with neat sketch. (CO1)

OR

- (b) With suitable sketch, explain the process of TIG welding. What are its disadvantages ? (CO1)

3. (a) With neat labelled sketch, explain Plasma Arc Welding. Write its advantages and disadvantages. (CO2)

P. T. O.

(2)

OR

(b) Describe the characteristics of underwater welding. Discuss some common problems encountered in underwater welding. (CO2)

4. (a) Compare laser beam welding and electron beam welding process with context of working principle, special characteristics, constructional features and applications. (CO2)

OR

(b) Explain the working principle of electron gun in electron beam welding. (CO2)

5. (a) With neat labeled sketch, explain the working of Ultrasonic Welding. (CO2)

OR

(b) Describe the different equipments needed for Explosive Welding. (CO2)

Roll No.

TAE-801

B. TECH. (AME) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
VEHICLE BODY SAFETY AND CONTROL

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define the terms 'active safety' and 'passive safety' in the context of automotive safety. (CO1)

OR

- (b) Explain the purpose of crumple zones in a vehicle and how they contribute to overall safety during a crash. (CO1)

2. (a) Discuss the types of impacts a vehicle may experience and how the design of crumple zones varies based on these impact types.

(CO1, CO3)

OR

- (b) Compare and contrast the characteristics of energy absorption in vehicle structures during a crash, emphasizing the importance of balance between stiffness and toughness. (CO1, CO3)

P. T. O.

(2)

3. (a) Propose innovative ideas for improving the behaviour of specific body structures in crash testing to enhance overall vehicle safety. (CO2)

OR

- (b) Define the concept of occupant safety in the context of vehicle crash testing. (CO2)

4. (a) Explain the importance of passenger compartment integrity during a crash and its role in protecting occupants. (CO4, CO3)

OR

- (b) Evaluate the effectiveness of airbags in enhancing occupant safety and preventing injuries during a collision. (CO4, CO3)

5. (a) Analyze the challenges associated with ensuring the safety of pedestrians in the event of a vehicle crash. (CO3, CO4)

OR

- (b) Develop a comprehensive plan for enhancing occupant safety through improvements in seating design and technology. (CO3, CO4)

Roll No.

TRA-801

**B. TECH. (ME-RA)
(EIGHTH SEMESTER)**

MID SEMESTER EXAMINATION, March, 2024

AUTOMATIC CONTROL

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define the following terms used in control system : (CO1)
- (i) Controlled and Manipulated variables
 - (ii) Disturbances
 - (iii) Poles and Zeros
 - (iv) Transfer function and
 - (v) Characteristic equations

OR

- (b) What do you understand by the open-loop and closed-loop control systems? Discuss with examples and their respective limitations. (CO1)

P. T. O.

2. (a) Describe the following system responses in terms of the poles mapping on s-plane : undamped, overdamped, underdamped, and critically damped systems. (CO1 & CO3)

OR

- (b) Plot and write the expressions (with Laplace transform) for the following signals : step, ramp, sinusoidal, and impulse. (CO1, CO3)
3. (a) Find the solution $x(t)$ of the following differential equation :

(CO1, CO2)

$$\ddot{x} + 3\dot{x} + 2x = 0 ; \text{ where } x(0) = a \text{ and } \dot{x}(0) = b$$

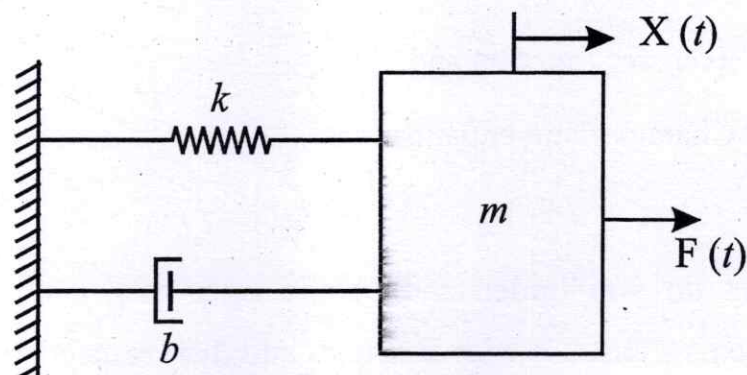
OR

- (b) Find the solution $x(t)$ of the following differential equation :

(CO1, CO2)

$$\ddot{x} + 2\dot{x} + 5x = 3 ; \text{ where } x(0) = 0 \text{ and } \dot{x}(0) = 0$$

4. (a) Derive the transfer function of the mass-spring-dashpot system as shown in the figure below. Also, comment about the stability and the behaviour of the system if, $m = 2$, $b = 1$, and $k = 10$: (CO3)



$$m = 2 \text{ kg}$$

$$b = 1 \text{ Ns/m}$$

$$k = 10 \frac{\text{N}}{\text{m}}$$

(3)

OR

- (b) What do you understand by the Routh's stability criterion ? Comment about the stability of a system, whose characteristic equation is given by the following polynomial : (CO3)

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

5. (a) Derive the expression for the unit-step response of a first-order system. Also, plot the response and discuss about the time constant. (CO2)

OR

- (b) Describe the following transient-response characteristics of a control system with a neat sketch : (CO2)

- (i) Delay time
- (ii) Rise time
- (iii) Peak time
- (iv) Maximum overshoot and
- (v) Settling time

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